

Session 12

1. Write equivalence partitioning (EP) test cases for the phone number input field in a Zomato-style food order form, where the field must accept exactly 10 digits and no letters or special characters.

Equivalence Partitioning (EP) – Zomato Phone Number

Requirement: Phone number must contain **exactly 10 digits**. Letters and special characters are not allowed.

Equivalence Partitions

Partition	Input	Valid/Invalid
EP1	Less than 10 digits	Invalid
EP2	Exactly 10 digits	Valid
EP3	More than 10 digits	Invalid
EP4	Contains letters	Invalid
EP5	Contains special characters	Invalid

Test Cases

TC ID	Test Data	Expected Result
TC01	987654321	Error – less than 10 digits
TC02	9876543210	Accepted – valid phone number
TC03	98765432101	Error – more than 10 digits
TC04	98765ABCD E	Error – letters are not allowed
TC05	98765@3210	Error – special characters are not allowed

Reasoning: EP divides all possible inputs into groups that are expected to behave similarly. We test one or more representatives from each group.

2. For a Flipkart-style sign-up form with a password field that must be 8-16 characters long, write boundary value analysis (BVA) test cases to check the minimum, just above minimum, maximum, and just above maximum password lengths.

Hint: Consider test cases for 7, 8, 9, 15, 16, and 17 characters.

Boundary Value Analysis (BVA) – Flipkart Password

Requirement: Password must be between 8 and 16 characters.

The given hint asks us to test 7, 8, 9, 15, 16, and 17 characters.

TC ID	Password Length	Example	Expected Result
TC01	7	Abc1234	Invalid – below minimum
TC02	8	Abc12345	Valid
TC03	9	Abc123456	Valid
TC04	15	Abc123456789012	Valid
TC05	16	Abc1234567890123	Valid
TC06	17	Abc12345678901234	Invalid – above maximum

Boundary Explanation

- 7 → just below minimum
- 8 → minimum allowed value
- 9 → just above minimum

- 15 → just below maximum
 - 16 → maximum allowed value
 - 17 → just above maximum
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3. Given an Instagram-like post creation form with a caption input that allows up to 2200 characters, use equivalence partitioning to identify valid and invalid input ranges and write at least 3 test cases for each partition.

Equivalence Partitioning – Instagram Caption

Requirement: Caption can contain a maximum of **2200 characters**.

Equivalence Partitions

Partition	Character Range	Valid/Invalid
EP1	0 characters	Valid
EP2	1–2200 characters	Valid
EP3	More than 2200 characters	Invalid

Test Cases – Valid Partition

TC ID	Input Length	Expected Result
TC01	0 characters	Caption can be left empty
TC02	100 characters	Caption accepted
TC03	2200 characters	Caption accepted

Test Cases – Invalid Partition

TC ID	Input Length	Expected Result
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TC04	2201 characters	Error – caption exceeds limit
TC05	2500 characters	Error – caption exceeds limit
TC06	3000 characters	Error – caption exceeds limit

Reasoning: Inputs are divided into valid and invalid groups. Testing representative values from each partition helps verify that the application handles the different input classes correctly.

4. For a BookMyShow movie ticket booking form with a 'number of tickets' field (allowed: 1 to 10), write BVA test cases to ensure the field correctly handles edge values.

BVA – BookMyShow Number of Tickets

Requirement: Number of tickets must be from 1 to 10.

For BVA, test the values around the minimum and maximum boundaries.

TC ID	Number of Tickets	Expected Result
TC01	0	Invalid – below minimum
TC02	1	Valid – minimum allowed
TC03	2	Valid – just above minimum
TC04	9	Valid – just below maximum
TC05	10	Valid – maximum allowed
TC06	11	Invalid – above maximum

Boundary Explanation

- 0 → just below minimum
- 1 → minimum
- 2 → just above minimum
- 9 → just below maximum

- 10 → maximum
 - 11 → just above maximum
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5. Pick any app you use daily (such as Paytm, Myntra, or IRCTC) and select one form field (e.g., amount, pincode, age). Write both equivalence partitioning and boundary value analysis test cases for that field, explaining your reasoning for each.

EP + BVA – Paytm Amount Field

Let's select **Paytm's payment amount field** as the example.

Assumption

For this assignment, assume the amount field accepts values from **₹1 to ₹50,000**.

This is a testing example for the assignment; actual limits can vary depending on the payment method and transaction rules.

A. Equivalence Partitioning

Partitions

Partition	Amount Range	Valid/Invalid
EP1	Less than ₹1	Invalid
EP2	₹1–₹50,000	Valid
EP3	More than ₹50,000	Invalid
EP4	Letters/text	Invalid
EP5	Invalid special characters	Invalid

EP Test Cases

TC ID	Test Data	Expected Result
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TC01	₹0	Invalid – amount must be at least ₹1
TC02	₹500	Valid amount
TC03	₹50,000	Valid amount
TC04	ABC	Invalid – letters not allowed
TC05	₹@500	Invalid – invalid characters

EP Reasoning

We divide the input into different groups. Values within the same group should produce the same type of result, so representative values from each group are tested.

B. Boundary Value Analysis

The allowed range is ₹1 to ₹50,000.

TC ID	Amount	Expected Result
TC01	₹0	Invalid
TC02	₹1	Valid – minimum
TC03	₹2	Valid – just above minimum
TC04	₹49,999	Valid – just below maximum
TC05	₹50,000	Valid – maximum
TC06	₹50,001	Invalid – above maximum

BVA Reasoning

BVA focuses on values **at and around the boundaries**, because defects commonly occur at minimum and maximum limits.

Final Short Conclusion

Equivalence Partitioning (EP) divides input data into valid and invalid groups and tests representative values from each group.

Boundary Value Analysis (BVA) tests values at the boundaries and immediately around them, such as **minimum -1, minimum, minimum +1, maximum -1, maximum, and maximum +1**.

These techniques help testers find input-validation defects efficiently without testing every possible input.